USN | | | | | | | | | | [23ESEC12]

# FIRST SEMESTER B.Tech EXAMINATION, JANUARY - 2024

## ELECTRICAL AND ELECTRONICS SYSTEMS

**Time: 03 HOURS** **Maximum Marks:100**

**Instructions:**
i. Missing data may be suitably assumed.
ii. Answer all questions.

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| | | **PART – A** | **05×02=10** | | |
| **1.** | **a** | Define frequency modulation. Write frequency modulated waveform with carrier and message signal. | **L2** | **CO1** | **02** |
| | **b.** | Define satellite. List different types of satellites. | **L2** | **CO2** | **02** |
| | **c.** | Explain the working principle of RADAR. | **L2** | **CO2** | **02** |
| | **d.** | Define renewable and non-renewable energy sources. | **L2** | **CO3** | **02** |
| | **e.** | Define power theft. How do you identify power theft? | **L2** | **CO4** | **02** |
| | | **PART - B** | **06×05=30** | | |
| **2.** | **a.** | Explain any five components of mobile phone. | **L2** | **CO1** | **05** |
| | | **OR** | | | |
| | **b.** | Explain any five components of monochrome television receiver. | **L2** | **CO1** | **05** |
| **3.** | **a.** | Explain the block diagram of satellite communication. | **L2** | **CO2** | **05** |
| | | **OR** | | | |
| | **b.** | List the advantages of wireless communication. | **L2** | **CO2** | **05** |
| **4.** | **a.** | Describe the advantages of internet of things. | **L2** | **CO2** | **05** |
| | | **OR** | | | |
| | **b.** | Explain the block diagram of AM transmitter. | **L2** | **CO1** | **05** |
| **5.** | **a.** | Illustrate the key principles of electric power transmission. | **L2** | **CO3** | **05** |
| | | **OR** | | | |
| | **b.** | List the advantages and disadvantages of electricity generation using nuclear energy. | **L2** | **CO3** | **05** |
| **6.** | **a.** | List the advantages of power transmission at high voltage with low current. | **L2** | **CO3** | **05** |
| | | **OR** | | | |
| | **b.** | Differentiate step up and step down transformer. | **L2** | **CO4** | **05** |
| **7.** | **a.** | Write a short note on smart grid. | **L2** | **CO4** | **05** |
| | | **OR** | | | |
| | **b.** | Describe the disadvantages of the existing electric grid. | **L2** | **CO4** | **05** |
| | | **PART - C** | **06×10=60** | | |

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|---|---|---|---|---|---|
| **8.** | **a.** | With the neat block diagram, briefly explain the operation of digital camera. | **L2** | **CO1** | **10** |
| | | **OR** | | | |
| | **b.** | With the neat block diagram, briefly explain the operation of smart watch. | **L2** | **CO1** | **10** |
| | | | | | |
| **9.** | **a.** | Explain the elements of wireless communication system. | **L2** | **CO2** | **10** |
| | | **OR** | | | |
| | **b.** | Briefly explain the block diagram of RADAR system. | **L2** | **CO2** | **10** |
| | | | | | |
| **10.** | **a.** | Illustrate any five RADAR applications. | **L2** | **CO2** | **10** |
| | | **OR** | | | |
| | **b.** | Explain layered architecture of IoT. | **L2** | **CO2** | **10** |
| | | | | | |
| **11.** | **a.** | With block diagram, explain how electricity is generated using steam. | **L2** | **CO3** | **10** |
| | | **OR** | | | |
| | **b.** | Describe the working principle of electricity generation using nuclear energy. | **L2** | **CO3** | **10** |
| | | | | | |
| **12.** | **a.** | Describe the working principle of hydro-electric power generation. | **L2** | **CO3** | **10** |
| | | **OR** | | | |
| | **b.** | Describe the construction of a transformer. | **L2** | **CO4** | **10** |
| | | | | | |
| **13.** | **a.** | Illustrate the key characteristics of smart grid. | **L2** | **CO4** | **10** |
| | | **OR** | | | |
| | **b.** | Briefly explain the functions of smart grid. | **L2** | **CO4** | **10** |